
MODEL SELECTION AND VALIDATION REPORT (remove_3_biomass model)

OBSERVATIONS:1719

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_3_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.45	0.39
remove_3_kde_fric	-0.09	0.03
remove_3_kde_feve	-0.12	0.03
sbt_mean	0.27	0.10
sbt_sd	-0.04	0.05
sst_sd	0.00	0.04
log_chla_mean	0.03	0.04
depth	-0.01	0.05
log_sum_fishing	0.11	0.03

Dispersion parameter: 0.85

Spatiotemporal AR1 correlation (rho): 0.95

Matérn range: 364.03

Spatiotemporal marginal AR1 SD: 1.12

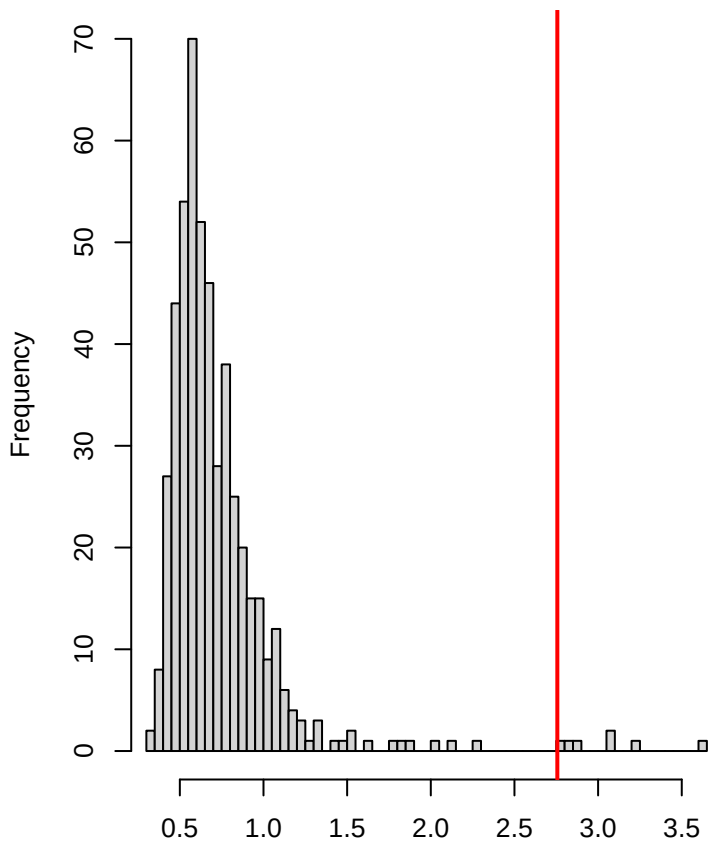
REML criterion at convergence: 12550.085

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

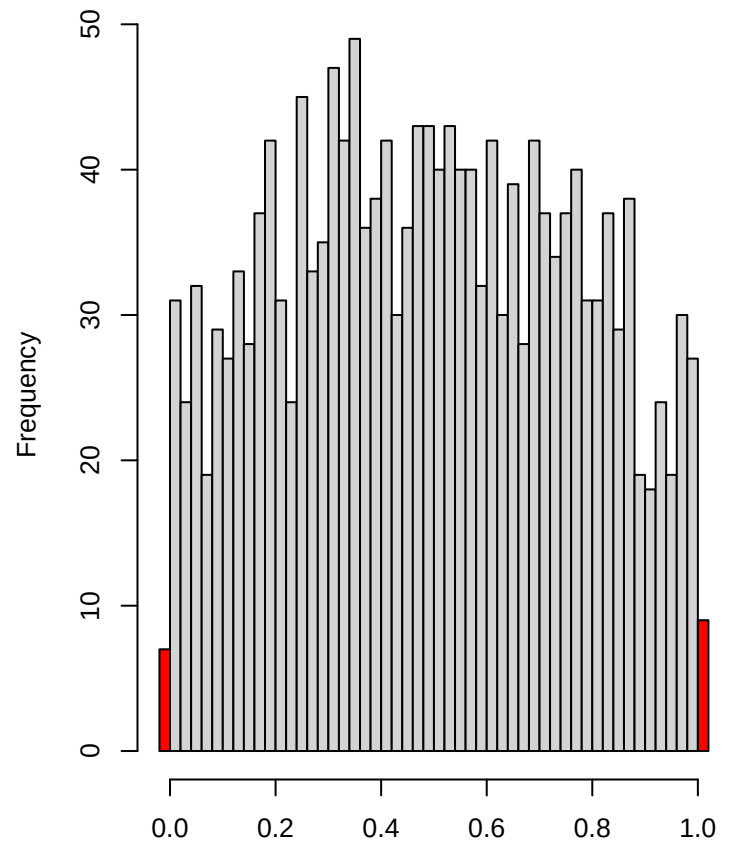
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	25126.17	2.7270778
spatiotemporal_random_fields_ar1_with_anisotropy	15	25128.90	155.2395521
spatial_random_fields	12	25284.14	0.1510922
spatial_random_fields_with_anisotropy	14	25284.29	43.3360984
spatiotemporal_random_fields_iid	12	25327.62	3.8897671
spatiotemporal_random_fields_iid_with_anisotropy	14	25331.51	293.5267232
error_only	10	25625.04	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



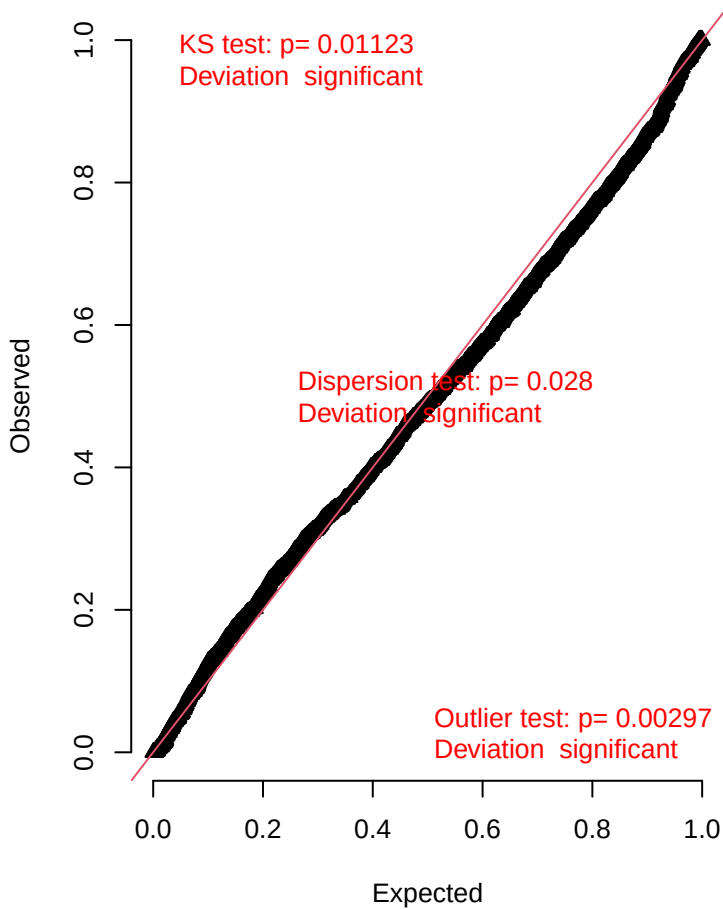
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

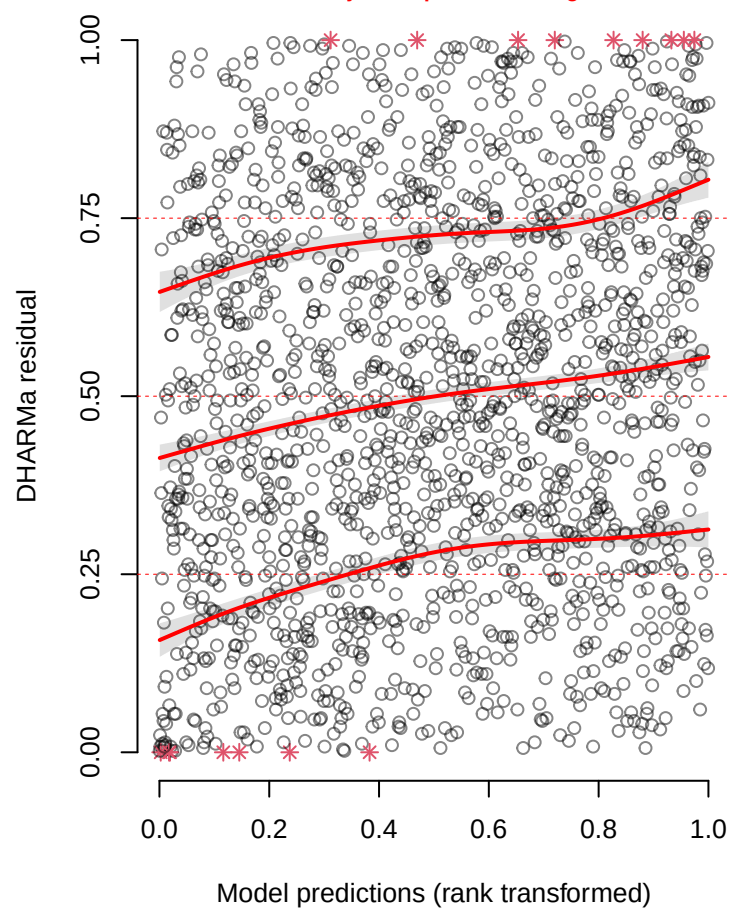


Residuals (outliers are marked red)

QQ plot residuals

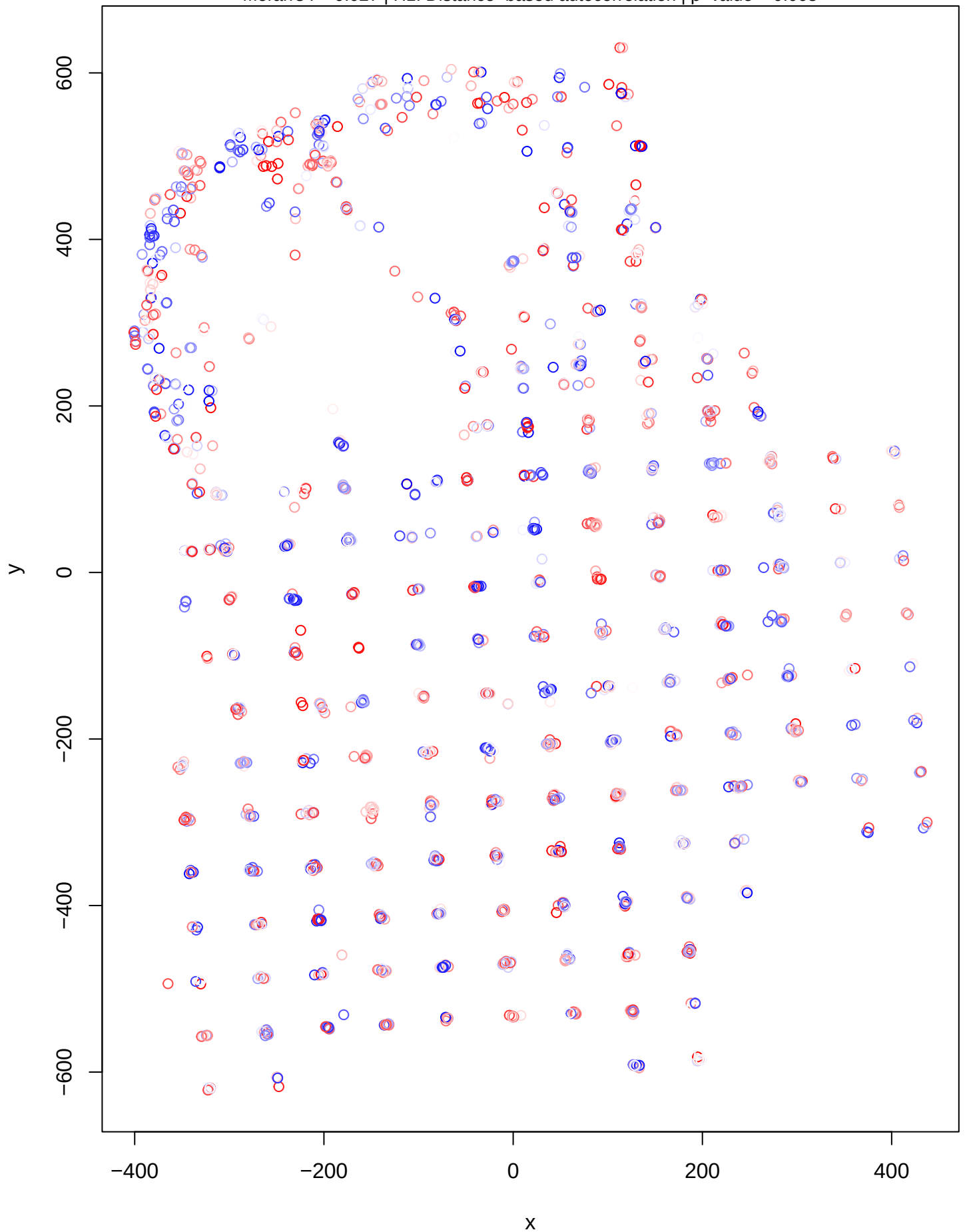


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant

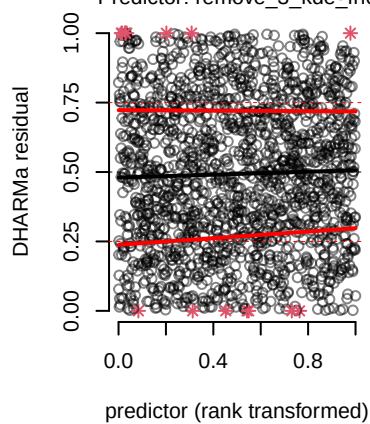


DHARMa Moran's I test for distance-based autocorrelation

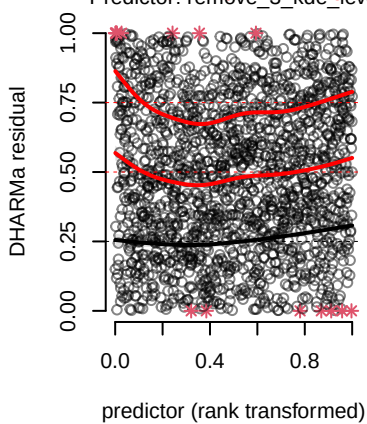
Moran's I = 0.027 | H1: Distance-based autocorrelation | p-value = 0.003



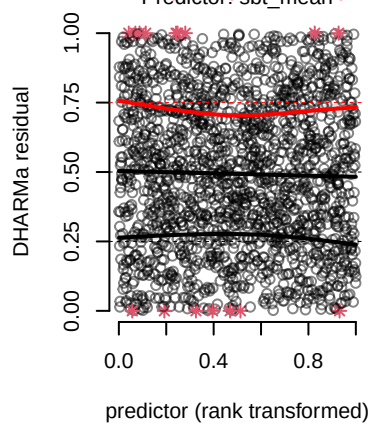
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



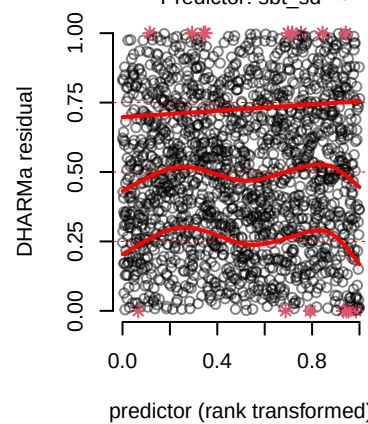
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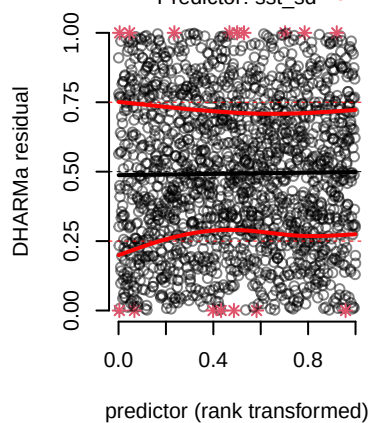
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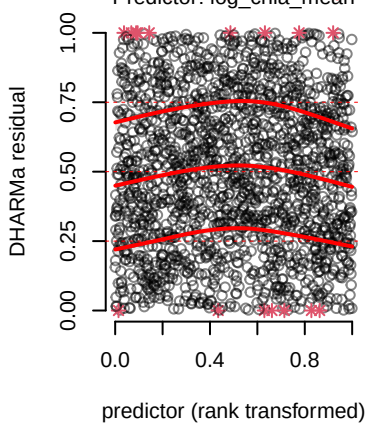
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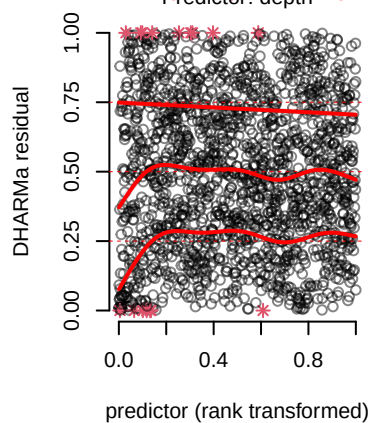
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